Neural Network Architecture

Introduction

An artificial neural network is a computational model inspired by the way the human brain processes information. It is made up of many simple processing units called neurons, which are connected to one another and organised in layers. By adjusting the strength of these connections, a neural network can learn patterns from data and use them to make predictions, classify inputs, or generate new content.

Layers of a Neural Network

The architecture of a basic neural network has three kinds of layers. The **input layer** receives the raw data, such as the pixels of an image or the numerical features of a record. Each neuron in this layer represents one input feature and passes its value forward without changing it. The **hidden layers** sit between the input and the output. These layers do most of the learning, because each neuron in them combines the values it receives and passes the result onward. A network with many hidden layers is called a deep neural network. The **output layer** produces the final result, for example a class label or a predicted number.

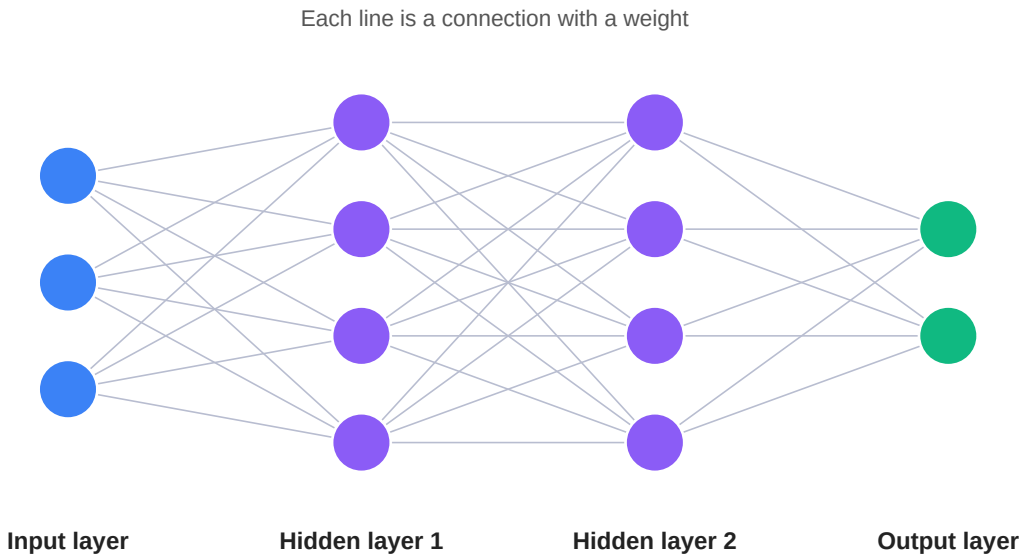

Figure 1: A simple feedforward neural network with two hidden layers

Weights, Biases and Activation Functions

Each connection between neurons has a **weight**, which controls how strongly one neuron influences the next. A neuron multiplies each incoming value by its weight, adds them together, and adds a **bias** term. The result then goes through an **activation function**, which decides how strongly the neuron fires. Common choices are ReLU, sigmoid, and tanh. Without activation functions, the network could only learn straight-line relationships, however many layers it had. The non-linearity they add is what lets networks learn complex patterns.

Training the Network

Training a neural network happens in two repeating steps. In **forward propagation**, the input passes through the layers to produce a prediction. A **loss function** then measures how far that prediction is from the correct answer. In **backpropagation**, the error is sent backward through the network, and an optimiser such as gradient descent adjusts the weights and biases to reduce it. Repeating this over many examples gradually improves the network's accuracy.

Common Architectures

Different tasks call for different architectures. A **feedforward network** passes information in one direction only and suits simple prediction problems. A **convolutional neural network (CNN)** uses filters to detect features such as edges and shapes, which makes it effective for images. A **recurrent neural network (RNN)** has connections that loop back, giving it a memory of earlier inputs, so it is used for sequences such as text and speech. **Transformers** use attention to relate every part of an input to every other part, and they now power most modern language models.

Conclusion

In conclusion, a neural network's design, meaning its layers, weights, activation functions, and connections, determines what it can learn. Choosing the right architecture for the problem is one of the most important steps in building a successful model.